

# Mastery Grids

1. What to focus on in this study session?



Improve on difficult concepts

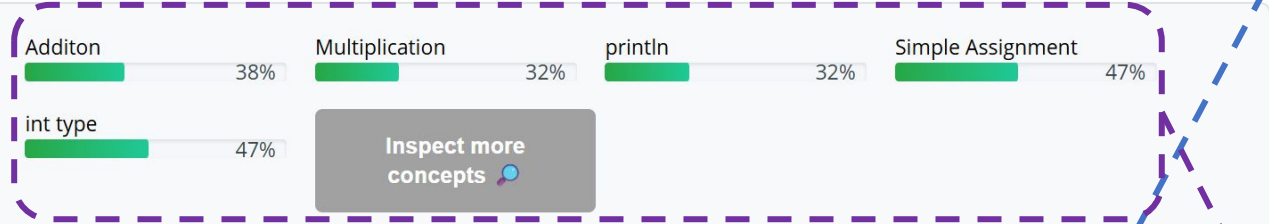


Work on concepts I haven't learned yet



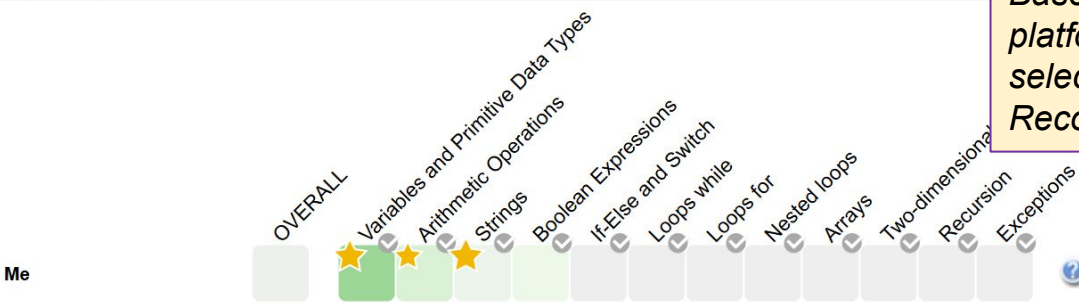
Catch me up with the class

2. Which concepts would you like to focus on?  
*Based on the learning goal selected, these priority concepts will be considered the focus for generating learning content recommendations.*



**If you have questions, contact:**  
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## My Progress



**Generation of personalized recommendations**  
*Based on estimated knowledge levels calculated by the system, the platform can suggest the most appropriate activities considering the selected **learning objective** and its **priority concepts**. Recommended activities are shown as **stars**.*

**Progress visualization**  
*The row ("Me") shows your personal progress (darker green means greater progress in that unit).*

Recommended Activities

1. Converting String to int

All Activities

Animated Examples

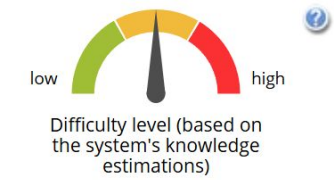
Tracing Problems

Programming Examples

Programming Challenges

Coding Problems





*Independent of its **adaptive learning support functions**, you can browse activities freely at **your own pace**.*

# How to increase your progress?

To increase your progress in the “Me” row (shown as *greener cells*), you need to interact with learning activities within each topic.

Click on a topic as shown below to access the content. Clicking the lines of “Explained examples” or correctly solving “Complete the code”, “Execution”, “Reorder the code”, or “Coding” activities will increase your progress percentage in each topic.

# Explained examples – Complete the code

Learn how a program is built line by line in “Explained examples” and challenge your understanding with “Complete the code”, where you must choose the missing lines to solve the given problem.

Example: Celsius To Fahrenheit Conversion

Construct a program that computes the Fahrenheit equivalent of an input Celsius value using the formula  $F = (9/5)C + 32$ . The input Celsius value is an integer.

```
1 import java.util.Scanner;
2 public class CelsiusToFahrenheit
3 {
4     public static void main (String[] args)
5     {
6         //Step 1: Define the data that we need for this program
7         final int BASE = 32;
8         final double CONVERSION_FACTOR = 9.0 / 5;
9         //Step 2: Read the input Celsius value
10        Scanner scan = new Scanner(System.in);
11        System.out.println("Enter the Celsius value: ");
12        int celsiusTemp = scan.nextInt();
13        scan.close();
14        //Step 3: Compute the Fahrenheit equivalent of the Celsius value
15        double fahrenheitTemp = celsiusTemp * CONVERSION_FACTOR + BASE;
16        System.out.println("Celsius Temperature: " + celsiusTemp);
17        System.out.println("Fahrenheit Equivalent: " + fahrenheitTemp);
18    }
19 }
```

Explanations

100% PROGRESS

NEXT

This line declares a constant named BASE and assigns it the value 32, which is the constant part of the formula for converting Celsius to Fahrenheit.

PREVIOUS

ADDITIONAL DETAILS

provide feedback on this explanation (optional)

Topic: Variables and Primitive Data Types • Activity: Fahrenheit to Celsius Conversion

Challenge: Fahrenheit to Celsius Conversion

Construct a program that computes the Celsius equivalent of an input Fahrenheit value using the formula  $C = (F - 32) (5/9)$ . The input Fahrenheit value is an integer.

Drag a tile to each missing field to construct this program.

```
1 import java.util.Scanner;
2 public class FahrenheitToCelsius {
3     public static void main(String[] args) {
4         //Step 1: Define the data that we need for this program
5         final int BASE = 32;
6
7         //Step 2: Read the input Fahrenheit value
8         Scanner scan = new Scanner(System.in);
9         System.out.println("Enter the Fahrenheit value: ");
10        int fahrenheitTemp = scan.nextInt();
11        scan.close();
12        //Step 3: Compute the Celsius equivalent of the Fahrenheit value
13
14        System.out.println("Fahrenheit Temperature: " + fahrenheitTemp);
15        System.out.println("Celsius Equivalent: " + celsiusTemp);
16    }
17 }
```

Drag a tile from here

CHECK

final double CONVERSION\_FACTOR = 5.0 / 9;

final double CONVERSION\_FACTOR = 5 / 9;

final double CONVERSION\_FACTOR = 9.0 / 5;

double celsiusTemp = fahrenheitTemp \* CONVERSION\_FACTOR - BASE;

double celsiusTemp = (fahrenheitTemp - BASE) \* CONVERSION\_FACTOR;

double celsiusTemp = fahrenheitTemp - BASE \* CONVERSION\_FACTOR;

## My Progress

Me

OVERALL

Variables and Primitive Data Types

Arithmetic Operations

Strings

Boolean Expressions

If-Else and Switch

Loops while

Loops for

Nested loops

Arrays

Two-dimensional Arrays

Recursion

Exceptions

Recommended Activities

2. Hello Printer

3. Multiple Variables

8. Constant Demo

9. Cash Register Constant Demo

All Activities

Animated Examples

Tracing Problems

Programming Examples

Programming Challenges

Coding Problems

Topic: Variables and Primitive Data Types • Activity: Sin

Tester

```
public class Tester{
    public static void main(String[] args) {

        int result;
        int i = 15;
        result = i * i;

    }
}
```

What is the final value of result?

Submit

Execution

Predict the result of running the presented program. This may be the console output (when it asks “What is the output?”) or the final value of the variable **result** (when it asks “What is the final value of result?”).

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## Reorder the code (Parsons problems)

Reorder the program’s code lines to solve the task given in the header. When your answer is complete, click “**Check**” to verify it or “**Reset**” to start over. **Pay close attention to indentation.**

tópico: Listas • Actividad: Actualizar elementos necesarios en una nueva lista

Construye un programa que encuentre los numeros pares en la lista original, calcule el cuadrado de cada uno de esos numeros y agregue los cuadrados a una nueva lista. El programa debe imprimir la nueva lista.

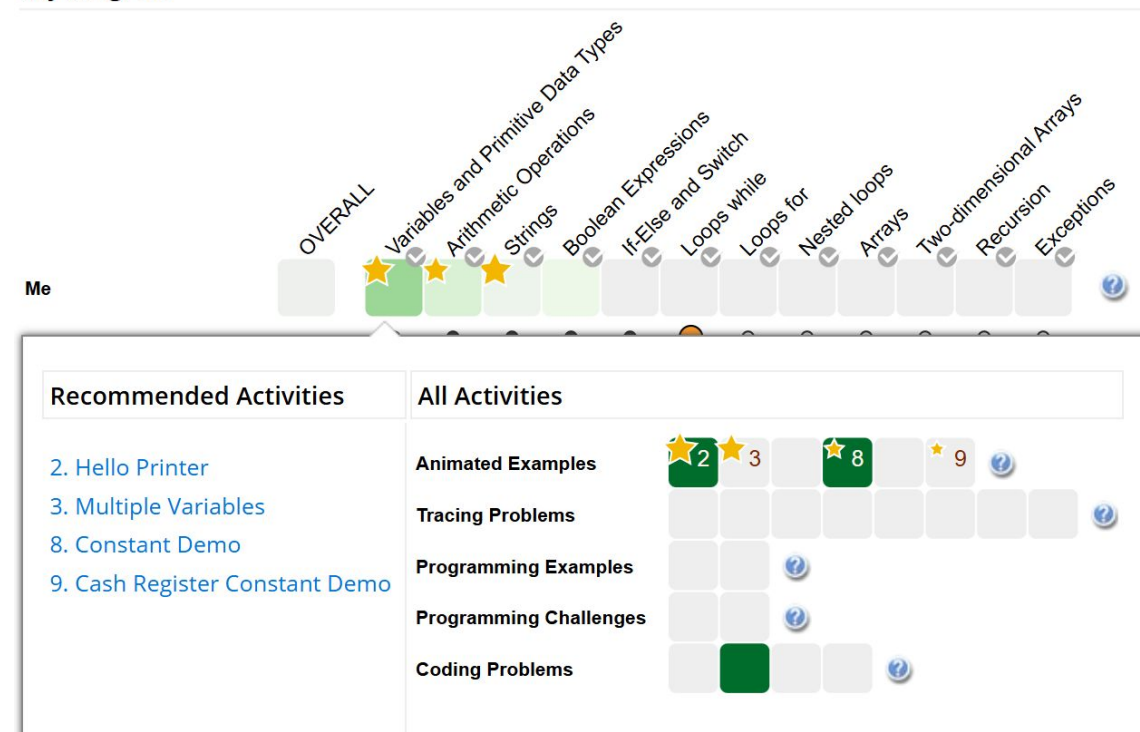
Drag from here

```
nueva_lista = []
nueva_lista.append(numero * numero)
numero = lista1[i]
print(nueva_lista)
```

Construct your solution here

```
lista1 = [1, 2, 3, 4, 5]
for i in range(len(lista1)):
    if numero % 2 == 0:
```

### My Progress



Topic: Variables and Primitive Data Types • Activity: Calculating the Perimeter of a Rectangle

Calculating the perimeter of a rectangle

Complete the following code in order to calculate the perimeter of a rectangle, given its height and width. The height and width of the rectangle are stored in the variables height and width, respectively. The initial value of both variables is already set to a positive number.

E.g. 1: if the value of height is 3 and the value of width is 4, the code prints 14.  
E.g. 2: if the value of height is 6 and the value of width is 7, the code prints 26.

```
1 int perimeter;
2 // TODO: add your code here
3 perimeter = height * 2 + width * 2;
4
5 System.out.println(perimeter);
```

✓ Your submission is correct!

Feedback

Passed

How difficult has this activity been for you? Easy Medium Hard

Close window

## Coding

Implement the code that solves the described problem. When you **submit the code**, it shows how many unit tests passed (happy face) and which failed (sad face). If all tests pass, the exercise is considered solved. **Your code progress is not saved, so remember to save your work.**